THE PRODUCT OF TWO REPRESENTABLE 1-MORPHISMS IS A REPRESENTABLE 1-MORPHISM

THE STACKS PROJECT AUTHORS

Editorial note. The source calls the result a formal consequence of the two stability lemmas, but the factorization and the base-change identifications are carried out in-span and are the load-bearing content. The two same-chapter inputs are not carded (both one-line 'entirely formal' proofs) and are declared external.

Allowed standard prerequisites (editorial).

- Composition stability of representability. lemma-composition-representable-transformations (same chapter, NOT carded, declared external). Composes the two base changes.
- Base-change stability of representability. lemma-base-change-representable-transformations (same chapter, NOT carded, declared external). Keeps each factor representable.

Source reference. Original lemma 02ZT: sources/stacks-algebraic/pinned-original-source.tex, source lines 525–525. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 02ZS: sources/stacks-algebraic/pinned-original-source.tex, source lines 508–508. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let S be a scheme contained in Sch_{fppf}. Let $\mathcal{X}_i, \mathcal{Y}_i$ be categories fibred in groupoids over $(Sch/S)_{fppf}$, $i = 1, 2$. Let $f_i : \mathcal{X}_i \to \mathcal{Y}_i$, $i = 1, 2$ be representable 1-morphisms. Then*

$$f_1 \times f_2 : \mathcal{X}_1 \times \mathcal{X}_2 \longrightarrow \mathcal{Y}_1 \times \mathcal{Y}_2$$

is a representable 1-morphism.

Proof. Write $f_1 \times f_2$ as the composition $\mathcal{X}_1 \times \mathcal{X}_2 \to \mathcal{Y}_1 \times \mathcal{X}_2 \to \mathcal{Y}_1 \times \mathcal{Y}_2$. The first arrow is the base change of f_1 by the map $\mathcal{Y}_1 \times \mathcal{X}_2 \to \mathcal{Y}_1$, and the second arrow is the base change of f_2 by the map $\mathcal{Y}_1 \times \mathcal{Y}_2 \to \mathcal{Y}_2$. Hence this lemma is a formal consequence of Lemmas 02ZS and 02ZT. □

Date: stacks-project:algebraic.tex:local-snapshot-20261012; GNU FDL 1.2 or later.